gets hotter and hotter as matter and radiation are compressed together more and more tightly by the shrinking of space. If we head back to a mere ten millionths of a second after the beginning, the universe gets so dense and so hot that ordinary matter disintegrates into a primordial plasma of nature's elementary constituents. And if we continue our journey, right back to nearly time zero itself—the time of the *big bang*—the entire known universe is compressed to a size that makes the dot at the end of this sentence look gargantuan. The densities at such an early epoch were so great, and the conditions were so extreme, that the most refined physical theories we currently have are unable to give us insight into what happened. For reasons that will become increasingly clear, the highly successful laws of physics developed in the twentieth century break down under such intense conditions, leaving us rudderless in our quest to understand the beginning of time. We will see shortly that recent developments are providing a hopeful beacon, but for now we acknowledge our incomplete understanding of what happened at the beginning by putting a fuzzy patch on the far left of the cosmic spacetime loaf—our verson of the terra incognita on maps of old. With this finishing touch, we present Figure 8.10 as a broad-brush illustration of cosmic history.

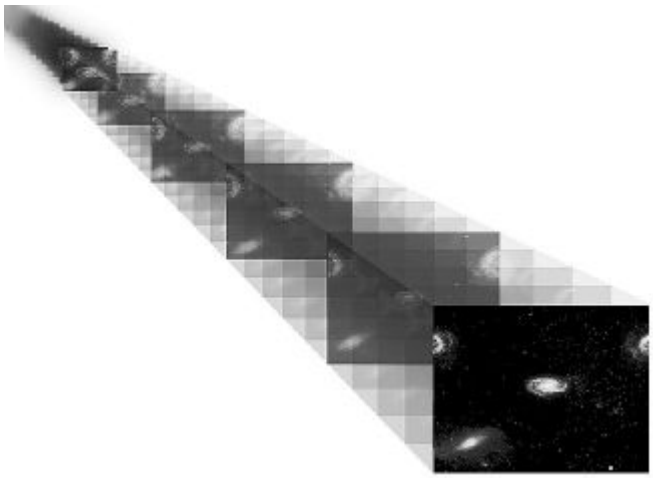


Figure 8.10 Cosmic history—the spacetime "loaf"—for a universe that is flat and of finite spatial extent. The fuzziness at the top denotes our lack of understanding near the beginning of the universe.

Alternative Shapes